

Strain response of C/SiC composite to thermal and mechanical load cycling in oxidising atmosphere

H. Mei* and L. F. Cheng

Strain response of a C/SiC composite simultaneously subjected to thermal cycling (1173–1473 K) and mechanical cycling (60 ± 20 MPa) in an oxygen–argon mixture: 10.4%O₂–89.6%Ar was investigated. Thermal strain with maximum value of $\sim 0.117\%$ and mechanical strain with maximum value of $\sim 0.029\%$ were directly measured on a progressively increasing baseline strain from 0.354 to 0.670%. The thermal strain and mechanical strain were found to have an approximately linear relationship with the temperature and stress respectively; and to be in quantitative agreement with the calculated results using the proposed semiempirical formulas. The baseline strain was considered to be damage dependent and could increase with accumulated damage. It is observed that the thermal and mechanical load cycling in an O₂–Ar mixture could cause the coating/matrix cracks, the degradation of the interfaces, fibre oxidation, and an increase in the baseline strain of the composites by widening crack opening displacement due to larger interfacial debonding and fibre sliding, and reducing effective load bearing area of the fibres.

Keywords: Ceramic matrix composites, Fatigue, Thermal cycling, Microstructure

Introduction

Continuous carbon fibre reinforced SiC ceramic matrix composites (C/SiCs) have undergone extensive development as candidate materials for structural applications during the past decade, due to their excellent thermo-mechanical properties and their capability to withstand both thermal and mechanical loads under severe conditions.^{1–6} In many applications, mismatch of the coefficient of thermal expansion (CTE) between fibre and matrix causes thermal stress which becomes the main reason for significant matrix cracking. Therefore, thermal cycling often causes permanent microstructural changes and significant strain increase in the serviced composite as well as the significant mechanical load cycling.

Previous efforts have been made to investigate the effect of either thermal cycling (e.g. thermal shock)^{7–12} or mechanical cycling (e.g. fatigue)^{3,13–19} on the damage development and resistance of the tested composites in oxidising environments, mainly based on their microstructural evolutions and mechanical degradation. However, strain response of the composites to both thermal and mechanical load cycling in oxidising atmosphere has not been experimentally reported yet, and in the combined environments mentioned above no description involving theoretical analysis and

calculation of strain response can be found in the recent literature.

In the present paper, experimental results on the strain response of a three-dimensional (3D) C/SiC matrix composite under thermal and mechanical load cycling in O₂–Ar mixture are presented. The thermal strain, mechanical strain and baseline strain were measured in real time and discussed. The microstructures were observed and the damage mechanisms were suggested.

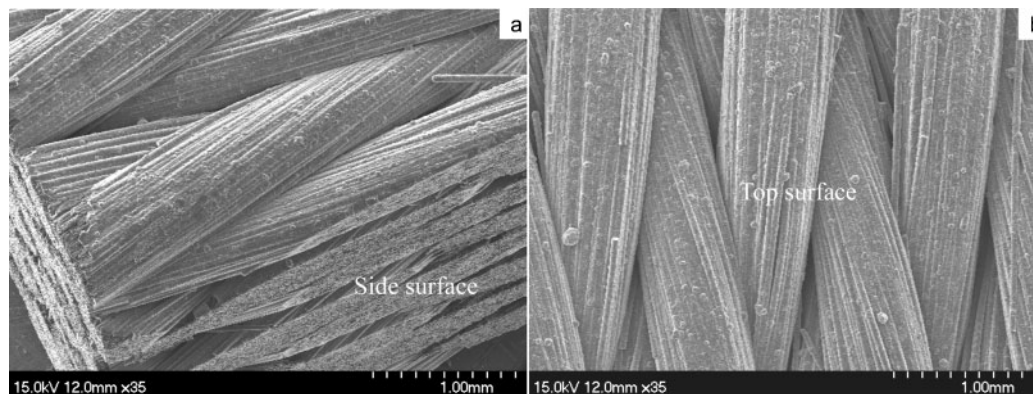
Experimental

Preparation of C/SiC composite

As described in a previous work,²⁰ 3D preforms were braided by a four step method using 1K T-300 carbon fibres. Composites were prepared by isothermal chemical vapour infiltration (CVI) technology at 1273 K. Volume fraction of fibres was $\sim 40\%$ and braiding angle was $\sim 20^\circ$. Methyltrichlorosilane (MTS, CH₃SiCl₃) was used for the infiltration of the SiC matrix with hydrogen as carrier gas. Typical conditions for deposition were 1273 K, a H₂:MTS ratio of $\alpha=10$, and a pressure of 4 kPa. Finally, the dog bone shaped test specimens were cut from the fabricated composite plates and further coated with SiC by I-CVI under the same conditions (thickness ≈ 50 μm). Fibre architectures of the materials are shown in Fig. 1 (Ref. 20) and dimensions of the specimens are $185 \times 3 \times 3$ mm. Major properties of the as received specimens are listed in Table 1.²⁰

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a 3D view; and b top view (Ref. 20)

1 Fibre architectures of as fabricated C/SiC composite

Thermal and mechanical load cycling tests

Thermal cycling experiments were conducted in an oxygen–argon mixture of 10.4 vol.-%O₂–89.6 vol.-%Ar using a newly developed integrated system, which was indicated in Fig. 2. Seven major critical points in this system were described in detail in the previous work.²⁰ As illustrated in the second right hand *y* axis in Fig. 3, thermal cycling was carried out between $T_{\text{lower}} \approx 1173$ K and $T_{\text{upper}} \approx 1473$ K over a period of 120 s. Nevertheless, due to thermal inertia the actual temperature cycle was a triangular-like wave, whose temperature range was ~ 300 K, and the mean heating/cooling rates were ~ 5 and -10 K s⁻¹ respectively. During the testing, a small fatigue stress of 60 ± 20 MPa (sinusoidal wave at a frequency of 1 Hz, stress ratio $R=0.5$) was simultaneously applied to the ends of the dog bone samples. The strain was measured directly from the gauge length of the samples by a contact Instron extensometer (Model A1452-1001B). Microstructures of the samples were observed with a scanning electron microscope (SEM, Hitachi S-4700).

Results and discussion

Strain response and analysis

Following the test procedure described previously, a strain response curve of the tested 3D C/SiC composite sample to cyclic temperature and fatigue stress was obtained, as shown in Figs. 3 and 4. As a representative example, the measured strain response data within the no. 18th thermal cycle were summarised in Table 2. It can also be seen from Fig. 3 that thermal strain of the tested sample almost linearly increases from 0.594% upon heating, then reaches a peak of 0.711% at the upper limit of 1473 K, and finally decreases to 0.601% with cooling back to the lower limit of 1173 K. The thermal strain difference between two temperature limits approached 0.117% during heating and 0.110% during cooling. As thermal cycle proceeded, the triangular-like thermal strain wave with the fixed maximum strain

range of 0.117% was repeated periodically as the same period of 120 s as the temperature cycle, although the base point strain was continuously increasing (Fig. 4). The calculated thermal strain rates are 1.95×10^{-5} s⁻¹ for heating and -3.67×10^{-5} s⁻¹ for cooling by linear fitting of the strain time curve in the Fig. 3. The thermal strain rate for cooling is approximately twice (≈ 1.9) that for heating. It is commonly accepted that the thermal strain has an absolute temperature dependence.^{20–24} Thermal strain and its rate can be expressed as

$$\varepsilon_{\text{Th}}(t) = \alpha_c T(t) \quad (1)$$

$$\dot{\varepsilon}_{\text{Th}}(t) = \frac{\partial \varepsilon_{\text{Th}}(t)}{\partial t} = \alpha_c \frac{\partial T(t)}{\partial t} = \alpha_c \dot{T}(t) \quad (2)$$

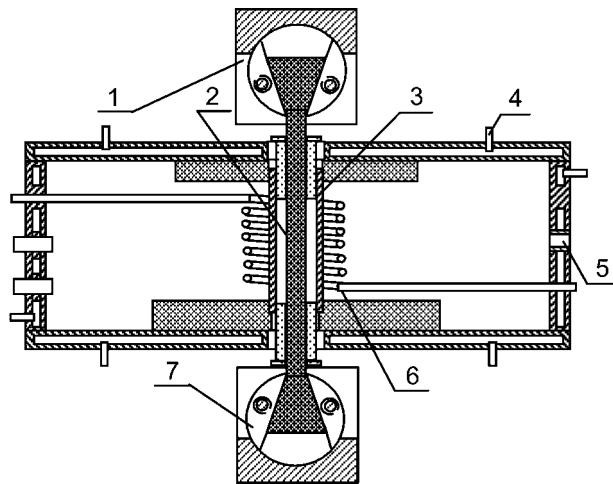
where, $\dot{T}(t)$ is the heating/cooling rate, α_c the CTEs along the composite material axis direction. Obviously, thermal strain and its rate are linear with T and $\dot{T}$ respectively. As described in the section on ‘Thermal and mechanical load cycling tests’, the mean heating and cooling rates were 5 and -10 K s⁻¹, and the latter was just twice the former. According to equation (2), the thermal strain rate during cooling should also be double that during heating. It is therefore reasonable that the ratio of the above observed thermal strain rate during cooling to heating was round 1.9. Furthermore, when the composite was thermally cycled, the thermal strain range $\varepsilon_{\text{Th}}^R$ between the lower and upper temperature limits at each cycle was determined as

$$\varepsilon_{\text{Th}}^R = \varepsilon_{\text{Th}}^{\text{upper}} - \varepsilon_{\text{Th}}^{\text{lower}} = \bar{\alpha}_c \Delta T \quad (3)$$

$\bar{\alpha}_c$ is the mean CTE between two selected temperatures, and ΔT is temperature difference. In Table 1, the mean CTE is 4.0067×10^{-6} K⁻¹ between 1173 and 1473 K, and $\Delta T \approx 300$ K. By equation (3) the thermal strain range should be 0.120%, which is in good quantitative agreement with the measured experimental value of 0.117%.

Table 1 Properties of as received 3D C/SiC composites (Ref. 20)

Property	Density, g cm ⁻³	Modulus, GPa	Strength, MPa	Porosity, %	CTE, $\times 10^{-6}$ K ⁻¹					
					973	1073	1173	1273	1373	1473
Value	2.1689	142.85	413.76	13.74	3.4198	3.8077	3.6415	3.7766	4.3293	4.3719



2 Schematic drawing of induction heating furnace with detailed view of grip holders, furnace and specimen (Ref. 20)

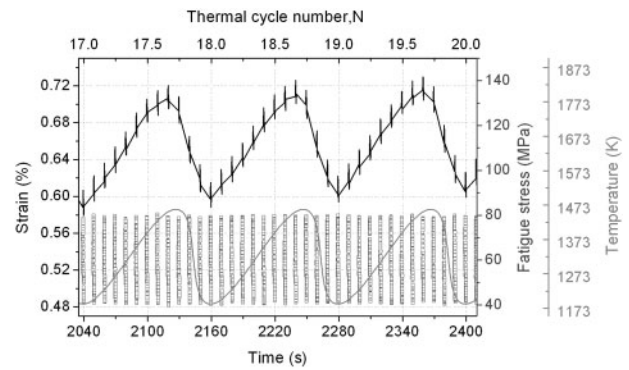
It can be noted that the fatigue stress of 60 ± 20 MPa also exhibits a significant influence on the total strain at different frequencies from the thermal cycling (Fig. 3). The fatigue strain data during testing were automatically recorded by the Instron tester every 10 s. It is actually observed in Fig. 3 that there are just 12 fatigue strain peaks appearing on each thermal strain wave within each period of 120 s, and these mechanical strains under 60 ± 20 MPa have nearly identical magnitudes, i.e. 0.024% for heating and 0.029% for cooling in average. Also as a representative example, the mean values of the measured 12 maximum–minimum fatigue strain peaks within the no. 18th thermal cycle are summarised in Table 2. In general, when the applied stress is below the so called proportional limit (i.e. the first matrix cracking stress $\sigma_{mc} \approx 50$ MPa for 3D C/SiC),²⁰ the mechanical strain of the composite is linearly elastic and dependent on the stress as the classical Hooke's law as

$$\varepsilon_{Me} = \frac{\sigma_{Me}}{E_c} \quad (4)$$

under a small fatigue stress, the mechanical strain range can be linearly obtained through

$$\varepsilon_{Fat}^R = \frac{\Delta\sigma_{Fat}}{E_c} \quad (5)$$

where, E_c is the Young's modulus and $\Delta\sigma_{Fat}$ is the fatigue stress range. In the present study, equation (5) can be utilised to simply assess the stress induced strain range, although the applied stress between 60 and 80 MPa ($\Delta\sigma_{Fat} \approx 40$ MPa) is slightly higher than the linear proportional limit of the tested 3D composite sample. As listed in Table 1, $E_c \approx 142.85$ GPa. By equation (5) the calculated fatigue strain range



3 Strain response curve of C/SiC composites to cyclic temperature and fatigue stress during thermal cycling ($\Delta T = 300$ K) and fatigue stress (60 ± 20 MPa)

$\Delta\varepsilon_{40}^R \approx 0.028\%$, which is ranked between the measured mean fatigue strain range of 0.029% during cooling and 0.024% during heating, and closer to the former. It is believed that thermal expansion of the composite sample during heating can weaken the increase in the mechanical strain in the same direction, whereas cooling shrinkage can enable the stress–strain relationship to basically follow the Hooke's law acting as a displacement constraint in an opposite direction.

An entire strain response curve of the C/SiC composite subjected to thermal cycling and mechanical fatigue is shown in Fig. 4. Figure 4b is a close-up of the initial strain cycles. Evidently, in the first cycle, the strain increased suddenly up to 0.471% upon heating to the upper temperature of 1473 K and then decreased to 0.354% with cooling back to 1173 K. The first heating from room temperature to 1173 K and the sudden applied stress of 60 ± 20 MPa must be responsible for the initial remarkable increase in the total strain. Through equations (1) and (4), the initial total strain can be written as

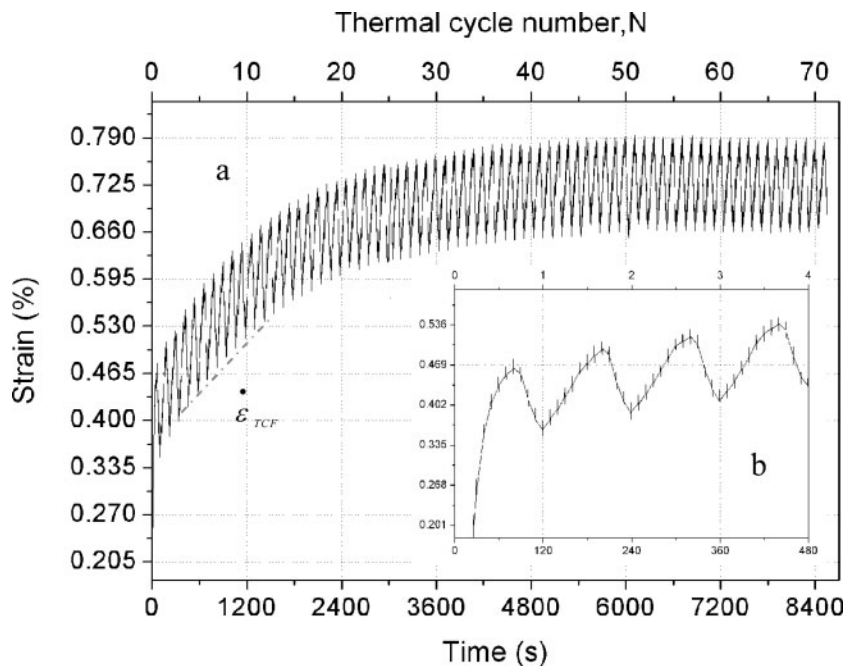
$$\varepsilon_0^{\text{Total}} = \varepsilon_0^{\text{Th}} + \varepsilon_0^{\text{Me}} = \alpha_c T_0 + \sigma_0 / E_c \quad (6)$$

Thus, $\alpha_c \approx 3.6415 \times 10^{-6} \text{ K}^{-1}$ at 1173 K in Table 1, $\sigma_0 \approx 60 \pm 20$ MPa and $E_c \approx 142.85$ GPa, the initial total strain should be $0.362 \pm 0.014\%$, which is in good agreement with the above observed value of 0.354%. The thermal strain is the largest part of equation (6) (i.e. $\varepsilon_0^{\text{Th}} \approx 0.320\%$) and accounts for 88.4% of the total strain. Therefore, thermal expansion resulting from heating is the dominant contribution to the total strain compared to the mechanical stress. Subsequently, the strain began to be cycled on the basis of the initial strain of 0.354% and the baseline strain continuously increased as thermal cycles proceeded. It is apparent from Fig. 4b that the total strain should be a combined result of thermal strain due to heating/cooling and mechanical

Table 2 Measured strain results of 3D C/SiC composites to thermal and mechanical load cycling within no. 18th thermal cycle

N	T, K	$\dot{T}$, Ks ⁻¹	$\dot{\varepsilon}_{Th}$, s ⁻¹	$[\varepsilon_{Th}^{\text{max}} - \varepsilon_{Th}^{\text{min}}]$, %	ε_{Th}^R , %	σ , MPa	$[\varepsilon_{Fat}^{\text{max}} - \varepsilon_{Fat}^{\text{min}}]^a$, %	ε_{Fat}^R , %
18th	Heating (1173–1473)	5	1.95×10^{-5}	0.711–0.594	0.117	60 ± 20	0.682–0.658	0.024
	Cooling (1173–1473)	–10	-3.67×10^{-5}	0.711–0.601	0.110	60 ± 20	0.663–0.634	0.029

$[\varepsilon_{Fat}^{\text{max}} - \varepsilon_{Fat}^{\text{min}}]^a$ gives the mean value of 12 maximum–minimum fatigue strains within no. 18th thermal cycle.



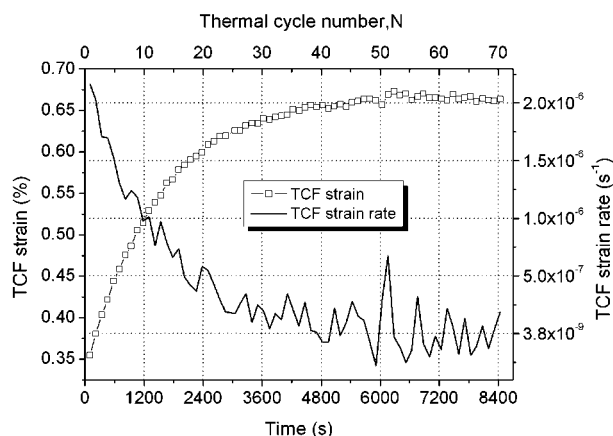
4 a strain curve of C/SiC composite simultaneously subjected to thermal cycling and fatigue stress in oxidising atmosphere and b close-up of strain within initial several cycles: slope of red broken line represents mean strain rate of thermal cycling fatigue

strain due to loading/unloading on a progressively increasing damage strain baseline. The baseline strain can be defined as 'thermal cycling fatigue strain' (hereafter, TCF strain) by

$$\varepsilon_{TCF} = \varepsilon_{T_{lower}}^{\sigma_{lower}} \quad (7)$$

$\varepsilon_{T_{lower}}^{\sigma_{lower}}$ refers to a retained strain of the tested sample at lower temperature limit and lower stress limit within each thermal cycle.

Figure 5 gives change in the TCF strain of the tested C/SiC composite and its rate when subjected to thermal cycles with $\Delta T \approx 300$ K and fatigue stress of 60 ± 20 MPa. It is noteworthy from Fig. 5 that the TCF strain increases from 0.354 to 0.670% at a decreasing rate (the right hand y axis of Fig. 5), and then levels off after ~ 50 cycles as the strain rate decreases to a negligible value ($3.8 \times 10^{-9} \text{ s}^{-1}$). It was indicated that the thermal cycling fatigue damage to the

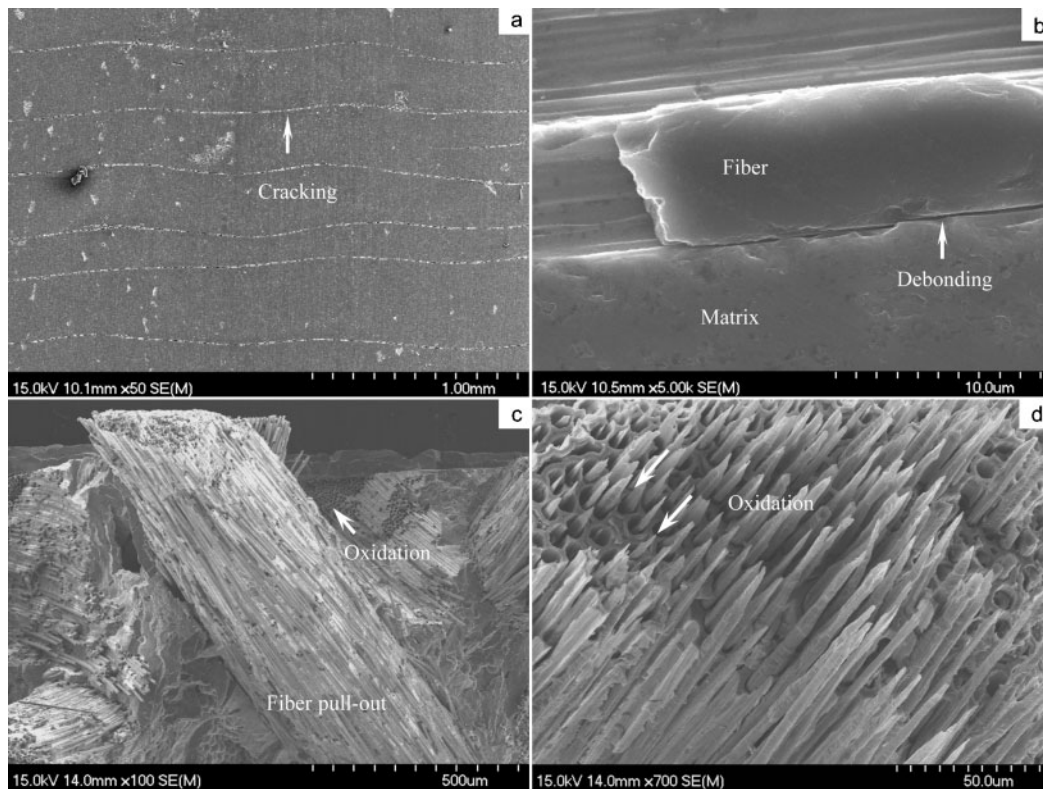


5 Change in TCF stain of C/SiC composites and its rate when subjected to thermal cycles ($\Delta T = 300$ K) and fatigue stress (60 ± 20 MPa)

composites mainly occurred before 50 cycles as a physical destructive mechanism, and diminished cycle by cycle with the destructive energy absorption via the formation the wider crack opening displacement owing to the longer interfacial sliding distance or more crack propagations. The formation of these damages prevented the composites from failing too early by increasing the TCF strain. This process continued to occur until an equilibrium point was reached, at which the composite was able to withstand all the applied loads, both thermal and mechanical, without further increase in the TCF strain. Subsequently, slow chemical recession (i.e. oxidation) of the carbon fibres in the composites could play a dominant role to continuously increase the TCF strain. Hence, non-linear increase in the TCF strain is ascribed to the result of the combination of the following two main contributing mechanisms:

- (i) physical damage mechanism, matrix microcracking normal to tensile axis, known as crack opening displacement (COD), accompanied by the interface debonding or sliding under thermal cycling and mechanical stress
- (ii) chemical damage mechanism, longitudinally increased compliance owing to reduction in the effective load bearing area resulting from the progressive oxidation of the reinforcing fibres from the surface to centre through the cracks created earlier.

As a consequence, the TCF strain can be acceptable to act as damage indicator when the composites are subjected to thermal cycling and mechanical stress in oxidising atmospheres. As shown in Fig. 4, the slope of the red broken line represents the mean TCF strain rate. The higher the TCF strain rate, the faster the TCF strain increased and the severer the damage to the composites. In a previous work,²⁵ the above two mechanisms can assist in modelling the effects of thermal, mechanical



6 Images (SEM) showing a cracks perpendicular to fibre axis, b fibre fracture and debonding, c long pullouts of fibres and bundles, and d oxidation of superficial fibres along propagating cracks beneath outer coatings

and chemical environments on damage evolution of the carbonaceous ceramic matrix composites on the basis of changes in real time baseline strain.

Microstructural observations

After 70 thermal cycles with the stress of 60 ± 20 MPa and in the O_2 -Ar mixture, microstructures of the cycled composites after the tension tests with a loading rate of 0.001 mm s^{-1} are presented in Fig. 6. Mean residual strength of the composite samples retained 89.1% of the as received properties. Figure 6a indicated that under thermal and mechanical load cycling, the SiC coating/matrix cracked at relatively regular transverse direction perpendicular to the axis fibre bundles in the composites. The coating and matrix cracks marked by arrows in the figures were partially closed once unloaded. In addition to the significant coating/matrix cracking, fibre debonding, bridging, and even fracture were easily found in the cycled composites (Fig. 6b). Thermal and mechanical cyclic reloading and unloading resulted in opening and closing of the matrix cracks, leading to repetitive shear stress (friction and wear) and slip of interfaces between fibre and matrix and thus long interfacial debonding resulted. The sliding resistance of interface τ_r can be estimated by¹³

$$\tau_r = \frac{\sigma_f d}{2l_c} \quad (8)$$

where σ_f is the fibre strength, d is the diameter of the carbon fibre ($\approx 7 \mu\text{m}$ as shown in Fig. 6b) and l_c is the critical length of the broken fibre. Normally, it is thought that the fibre pullout length is equal to $l_c/2$.

The thermal and mechanical load cycling could reduce the sliding resistance τ_r by weakening the interface. This

process increased the length of the debonded interface (Fig. 6b), resulting in long fibre pullouts when the fibre strength σ_f is assumed to be constant in equation (8). As shown in Fig. 6c, large quantities of long pullout fibres are observed in the fractured section of thermal cycled composites, and the pullout length of a representative bundle is considerably long ($\sim 1 \text{ mm}$). The 3D C/SiC composites exhibit a typical toughening and non-catastrophic failure characteristic.

On the other hand, the tested composites also suffered a slight oxidation. Figure 6d presents a close-up of the superficial fibres along the propagating coating cracks around the outer bundles on the fractured sections. In the presence of the mechanical loading, the fibre oxidation could result in the increase in the TCF strain of the composite by reducing the effective load bearing area. As illustrated in Fig. 6d, the oxidised fibres failed to hinder the propagations of the transverse cracks, leading to the loading transfer from the outside to inside fibres where the surviving fibres were now subjected to a larger tensile stress, increasing hence, their probability of failure and TCF strain as the above mechanism (ii).

Conclusions

Strain of a C/SiC composite subjected to thermal cycling (1173–1473 K) and mechanical fatigue (60 ± 20 MPa) in 10.4 vol.-% O_2 –89.6 vol.-%Ar atmosphere was investigated. By different frequency response characteristics of thermal and mechanical load cycling, thermal strain value of $\sim 0.117\%$ and mechanical strain value of $\sim 0.029\%$ were directly measured on a progressively increasing TCF strain from 0.354 to 0.670%. The thermal strain and mechanical strain are found to have

an approximatively linear relationship with the temperature and stress respectively; and to be in quantitative agreement with the calculated results using the proposed semiempirical formulas. The TCF strain is considered to be damage controlled, which depends mainly on the following two main contributing mechanisms: (i) formation of matrix microcracking accompanied by the interface debonding or sliding, and (ii) enhanced compliance of the test specimen owing to fibre oxidation. As confirmed by the microstructural observations, the thermal and mechanical load cycling in oxidising atmosphere could cause the coating/matrix cracks, the degradation of the interfaces, fibre oxidation, and an increase in the baseline strain of the composites by widening the crack opening displacement and reducing the effective load bearing area of the fibres.

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